Sky Polarisation Compass Aiding of Stereo Visual Odometry for GNSS Denied Desert Robots under Dust, Tilt and the Noon Blind Window

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***Abstract*—A ground robot that loses satellite positioning in a desert must keep its heading while its cameras degrade in dust, glare and low light. Desert ants use a sky polarisation compass for this. We test, in a third and stronger study, when such a compass helps stereo visual odometry and how a filter should weight it. A vector Monte Carlo radiative transfer sky with climatological aerosol, checked against measured desert values and one day of open full sky polarisation images, brackets the noon blind window of a zenith compass in Abu Dhabi on 21 June at 68 to 103 minutes and places such windows at every latitude within about 34 degrees of the equator. A Cramér Rao analysis shows that no polarisation geometry gives heading with the sun exactly at the zenith, while a ring of off zenith points shortens the window to minutes or less. With a compass bias, dust tied to the image visibility, a filter that no longer knows the true compass noise and a separate tilt sensitivity run, we compared nineteen systems under a protocol frozen before a confirmation run on 15 rendered trajectories from three environments never used before. A weighted compass lowered the heading RMSE from 5.10 to 1.98 degrees; an ungated compass was worse than none even with clear images; DoP floored covariance matching was non inferior to covariance matching; a ring geometry beat the zenith point on level poses but lost its advantage under a 2 degree tilt; and SIMULATED wheel odometry with a sky compass drifted 1.22 percent of path length at the most severe level against 3.46 for vision alone. A calibrated noise model did not repair the overconfidence, nor did oracle heading or translation, and the ranking among weighting schemes did not transfer between splits. Scenes are rendered and the compass is simulated.**

***Index Terms*—GNSS denied navigation, polarisation compass, radiative transfer, stereo visual odometry, desert robotics, adaptive Kalman filtering, path integration**

I. Introduction

Ground robots usually take their position from a global navigation satellite system (GNSS). When that signal is lost, dead reckoning from inertial and visual sensors has to carry the robot until it returns. The loss is not hypothetical: aircraft operators reported 37,770 events of GPS signal loss in the Middle East region in 2023, 92.96 per 1,000 flights against 29.83 worldwide, in data presented to ICAO [1]; these are reported events, not confirmed jamming. Aviation and maritime authorities describe jamming and interference in the Gulf [2]–[4], and the United Arab Emirates reported an increase in interference cases in 2025 [5].

A visual inertial fallback has two weaknesses in a desert. Global heading is unobservable from relative measurements [6], [7], and the camera itself degrades in sand, glare and dust [8], [9]. Desert ants solve a similar problem with a sky polarisation compass and a stride integrator [10], [11], [17], and robots built on that idea have homed over metres [15], [20]. We take from the ant one transferable idea, an absolute heading from the sky combined with a relative motion estimate, and ask when it helps a stereo visual inertial filter on a desert rover, and how the filter should weight it.

This paper reports the third study of a project whose first two studies used a surrogate front end (Study 1) and rendered desert scenes with a single scattering sky (Study 2). Study 2 found that an ungated sky compass is harmful around solar noon, that weighting by the measured degree of polarisation (DoP) or by innovations helps, and that all filters were overconfident. A red team review of Study 2 found that this rested on a sky without multiple scattering, a compass without bias or tilt, a weighting scheme matched to its own simulator, sky conditions independent of the image degradation, and a reused test set. Study 3 rebuilds the evidence so that each of these can fail.

Contributions. (i) A vector Monte Carlo radiative transfer sky with climatological aerosol, checked against measured desert DoP values [23] and against one day of open full sky polarisation images [67]; it brackets the noon blind window of a zenith compass in Abu Dhabi at 68 to 103 minutes on 21 June and places such windows at all latitudes within about 34 degrees of the equator (Section III). (ii) A Cramér Rao analysis showing that no sky polarisation geometry gives heading at the

instant the sun is at the zenith, while a ring of off zenith points shortens the window to 11 minutes or less. (iii) A compass error model with bias, exact tilt geometry and dust coupled to the image visibility, under which the filter no longer knows the true compass noise. (iv) Nineteen systems, including new robust, two mode, bias state and learned weighting schemes, wheel odometry, a sun sensor and a magnetometer, compared under a protocol frozen before a confirmation run on 15 trajectories from three environments never used before. (v) A simulated homing study and a drift decomposition. All scenes are RENDERED, all compass data SIMULATED; the only MEASURED inputs are the sky polarisation checks; the aerosol climatology is REANALYSIS.

II. Related Work

Insect inspired navigation. The ant toolkit comprises a skylight compass, a path integrator and view memory [17], [18]. Sahabot [20] and AntBot [15] navigated with polarisation compasses, and recent work pairs odometry with learned views [21], [22]; a review finds no study yet showing insect inspired navigation working robustly [16].

Skylight polarisation as a compass. Single scattering Rayleigh theory gives the e vector normal to the scattering plane [45]; measured patterns deviate under aerosol and cloud [23], [46], [53], neutral points move with turbidity [62], and vector radiative transfer with aerosol profiles reproduces measured skies [61]. Lower DoP means larger angle error [54]; near the zenith the solar azimuth is undefined and yaw is lost [55], [56]; tilt adds errors that grow with tilt and solar altitude [47], and vision based tilt compensation has been proposed [63]. Multi directional sensors combined with a sun sensor target cloudy skies [65], and a fused pattern compass was recently tested in dust [64]. Engineered sensors report sub degree to a few degrees of heading error [49], [50], [56], and a single camera compass already estimates its own covariance [66]. Open full sky polarisation image sets now exist [67].

Heading in visual inertial estimation and its weighting. Visual inertial filters [7], [24]–[26] are overconfident in yaw without observability aware design [6], [27], [28]. Polarisation compasses have been fused with visual inertial filters on micro aerial vehicles [59] and road vehicles [60]. For weighting, covariance matching [32], chi square based inflation [33], DoP thresholds [57], variational Bayes and innovation saturation [41], [58] have been used, mostly without a camera. Planetary analogue datasets (MADMAX [9], S3LI [68]) expose visual odometry to sand and glare.

Gap. We found no work, among those we opened, that couples a radiative transfer sky with dust to the camera degradation, weights a compass by the measured DoP inside visual odometry, and confirms the ranking on unseen environments under a frozen protocol. Each part has precedent, and we cite it; the combination and the confirmation are what this study adds.

III. Sky and Compass Model

A. Radiative transfer

The earlier studies used the single scattering Rayleigh law $d = d_{\max} \sin^2 \theta_s / (1 + \cos^2 \theta_s)$ for the zenith DoP, with $d_{\max} = 0.6$ [23], [47]. Study 3 replaces it with a vector Monte Carlo radiative transfer model (SIMULATED) of a plane parallel atmosphere: Rayleigh scattering (optical depth 0.17, about 470 nm), one aerosol type with a Henyey Greenstein phase function ($g = 0.7$, single scattering albedo 0.92) and a Rayleigh shaped linear polarisation of maximum 0.2, a Lambertian depolarising ground (albedo 0.3) and full multiple scattering [53], [61]. Stokes vectors are carried with explicit reference frames and the radiance toward the zenith and toward a ring of twelve points at 60 degrees elevation is estimated by the local estimate method. In the single scattering limit the code reproduces the analytic Rayleigh DoP and the zenith angle of polarisation to numerical precision. The aerosol phase matrix is a simplification, not a Mie computation; its parameters and the albedo are varied in a sensitivity run. The aerosol optical depth comes from the NASA POWER monthly climatology (MERRA-2 reanalysis, labelled REANALYSIS) at 550 nm for Abu Dhabi (annual mean 0.45, June 0.56, July 0.73) [69]; AERONET was not reachable from the analysis environment.

B. Validation against measurements

Two measurements constrain the model. In the Tunisian desert, Horváth and Wehner measured a zenith DoP of 24 to 30 percent at a solar elevation of 35 degrees and about 8 percent at 70 degrees [23]. The model gives 0.26 at that first geometry for the lowest aerosol (AOD 0.15) and 0.18 at the Abu Dhabi mean, and 0.021 at 70 degrees; the near sun value of 8 percent exceeds even single scattering Rayleigh (6.2 percent), so we treat it as unexplained. Second, we analysed one day (29 July 2022) of the open Marseille full sky polarisation images of Poughon *et al.* [67] (MEASURED, CC BY-SA 4.0): 382 daylight images, zenith patch of 80 by 80 pixels. The image timestamps were taken as local summer time (UTC+2), which also maximised the correlation of the zenith DoP with $\sin^2 \theta_s$ (0.98), so this correlation is a consistency check rather than independent validation, and on images labelled clear the Rayleigh law fits with $d_{\max} = 0.613$, essentially the value assumed in the earlier studies, while the radiative transfer model at low aerosol lies about 15 percent lower at solar zenith 50 to 70 degrees. The measured angle of polarisation follows the solar meridian with a robust standard deviation (1.4826 times the median absolute residual) of 6.4 degrees over all daylight images and 5.9 degrees on clear images, after fitting the unknown camera orientation; this residual includes the camera tilt and centring and is not used to refit the sensor. We therefore keep the radiative transfer table as the primary (conservative) model and add a measured anchored table, with the clear sky law from the Marseille fit and aerosol and dust attenuation from radiative transfer, as a secondary model. No

sign flip of the zenith polarisation (a neutral point) appears in any table cell with a usable DoP.

C. Blind window, geometry and latitude

With the frozen sensor law $\sigma_\alpha^2 = \sigma_f^2 + (c/d)^2$ ($\sigma_f = 0.5°$, $c = 0.05°$), the zenith compass noise on 21 June in Abu Dhabi exceeds 10 degrees for 103 minutes under radiative transfer with the June aerosol (97 to 108 minutes for half to one and a half times that aerosol) and for 68 minutes with the measured anchored table, against 64 minutes in Study 2; over half to one and a half times the June aerosol the two tables give 62 to 108 minutes, and differences of a few minutes are within the Monte Carlo resolution. At the mean climatological aerosol the window lies only in May, June and July on the 21st of each month examined; with one and a half times that aerosol August also has a short window. Fig. 1 shows the model and the day.

Heading information from a set of sky points follows from the Cramér Rao bound $\sigma_\psi^{-2} = \sum_k (\partial\alpha_k/\partial\psi)^2/\sigma_{\alpha,k}^2$, with the derivative of each point's angle of polarisation from the scattering plane geometry and its noise from the radiative transfer DoP. A Monte Carlo least squares estimator matches the bound to within 3 percent from 3 to 40 degrees solar zenith. At the instant the sun is exactly at the zenith the pattern is rotationally symmetric and no polarisation geometry can give heading; away from it a ring of points at 60 degrees elevation gains information much faster than the zenith point, because the off zenith angles rotate with the sun azimuth at first order while the zenith DoP grows only at second order. Adding the ring cuts the 21 June window to 11 minutes (radiative transfer) and 0 minutes (anchored); a sun direction sensor (0.5 degrees, ASSUMED) has a window of 23 minutes. Whether a ring sensor keeps any operational window in Abu Dhabi, where the sun passes about 1 degree from the zenith, therefore depends on the near zenith DoP; at higher sun or at a site where the sun crosses the zenith exactly, only a non celestial cue such as a magnetometer (ASSUMED error model) gives heading. Under radiative transfer a zenith compass window occurs at every latitude between about 34 degrees south and north, which includes the Sahara, the Arabian deserts, the Thar, the Kalahari, the Namib, the Atacama and the Australian deserts (Table I).

D. Compass error model

Study 3 adds, on top of the point noise: a heading bias (initial 1 degree, random walk 0.01 degrees per root second); the exact tilt error of a compass that assumes it is level, from the sun geometry and the platform roll and pitch (a Gauss Markov process of 2 degrees and 20 s on the development split; the protocol took roll and pitch from the rendered poses on the confirmation split, which turned out to be level, so tilt enters there through a sensitivity run); broken cloud with the POWER monthly cloud amount as the probability of a cloudy scenario; and, in the primary analysis, a near ground dust layer whose optical depth $3.912H/V$ ($H = 100$ m, assumed) follows the image visibility V of each degradation level, so that a 20 m visibility image comes with a dust sky. The filter's noise estimate uses only the measured zenith DoP and the frozen law, so it no longer matches the simulated error; this removes the circularity of the earlier B3 result.

TABLE I
LONGEST DAILY WINDOW (MINUTES WITH HEADING NOISE ABOVE 10 DEGREES) AND DAYS PER YEAR WITH A WINDOW, ZENITH POINT AND ZENITH PLUS RING, RADIATIVE TRANSFER SKY WITH AEROSOL 0.3 (SIMULATED).

Desert	Lat.	Zenith	Ring	Days
Thar	27	94	0	98
Abu Dhabi	24	97	11	118
Sahara, central	23	97	12	122
Rub al Khali	20	95	12	140
Sahara, Sahel edge	16	93	12	162
Great Sandy	-20	95	12	134
Kalahari	-23	97	12	118
Namib	-24	97	11	110
Atacama	-25	97	0	106
Simpson	-25	97	0	106

IV. SYSTEM

A. Data, degradation and visual odometry

All studies use TartanGround [38] (RENDERED, CC BY 4.0 [52]), the first 200 s of each trajectory, the front stereo pair at 320 by 320 pixels, and the frozen Study 2 degradation (Koschmieder dust haze along depth [51], low light with exposure and noise, motion blur, glare, frame drops; levels D0 to D4 with visibility none, 1000, 200, 60 and 20 m) and stereo visual odometry (KLT tracking, PnP with RANSAC, no mapping). Fig. 2 shows the levels. The IMU is the noise free dataset IMU with added noise and bias random walks.

B. Filter

A ten state extended Kalman filter estimates planar position, yaw, world velocity, gyro bias, two accelerometer biases, a compass heading bias and a visual forward speed scale error. The two new states are inert (zero variance) unless a system uses them, and a unit test checks that the filter then reproduces the Study 2 filter exactly. The IMU drives the prediction at 100 Hz; visual odometry updates body velocity and gyro bias at 10 Hz with bounded inflation of outliers; the compass updates yaw at 5 Hz.

C. Systems

Table II lists the nineteen systems. B0 to B3 are the earlier baselines with their frozen parameters. B4 (DoP floor for covariance matching, no gate) was exploratory in Study 2 and is tested here. B5 to B8 are new weighting schemes; one of them, S, is selected on the development split by a written rule. B3L adds a lock out recovery rule, B3C the calibrated noise model fitted on the pilot split (a Gauss Markov visual speed scale with $\sigma = 0.078$ and $\tau = 4.8$ s, and the variance of the averaged gyro in the bias measurement, which the earlier filter omitted). B3R and B4R use the zenith and ring geometry, SUN and MAG a sun sensor and a magnetometer with assumed error

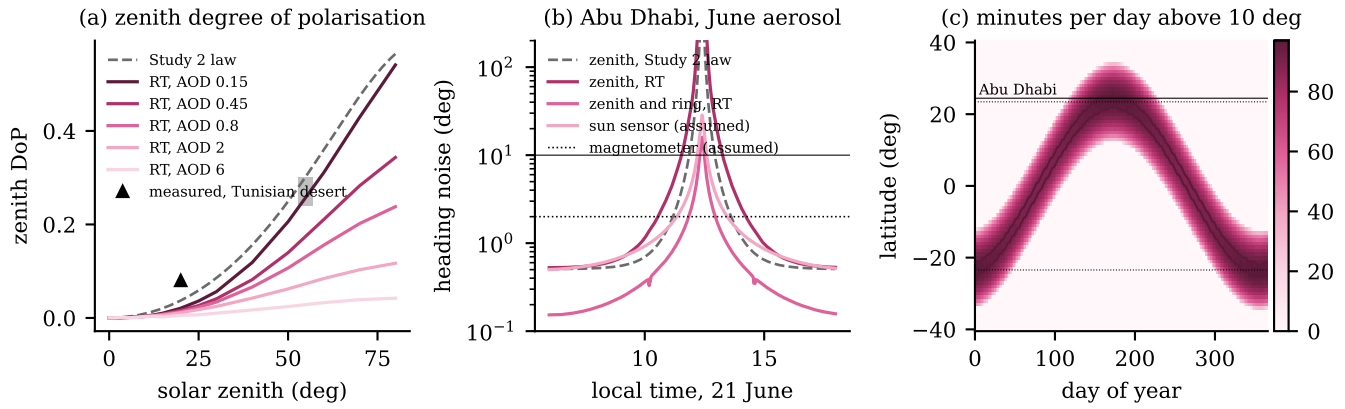


Fig. 1. Study 3 sky model (SIMULATED, with MEASURED reference points). (a) Zenith DoP against solar zenith for the radiative transfer model at five aerosol optical depths, the earlier single scattering law and the measured desert values [23] (grey box and triangle). (b) Heading noise on 21 June in Abu Dhabi with the June aerosol for the zenith point, the zenith plus ring geometry and a sun sensor; dotted line, the assumed magnetometer. (c) Minutes per day with zenith compass noise above 10 degrees by latitude and day of year (aerosol 0.3).

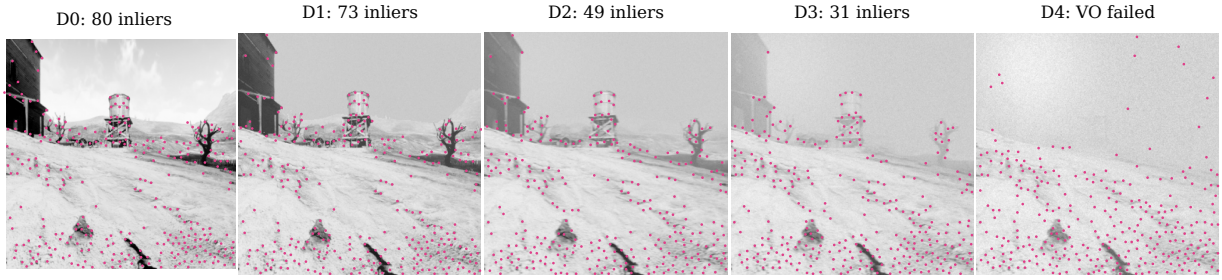


Fig. 2. One pilot frame at levels D0 to D4 (RENDERED image, SIMULATED degradation) with tracked features and PnP inliers.

TABLE II
STUDY 3 SYSTEMS (ONE FILTER, DIFFERENT COMPASS AND ODOMETRY INPUTS).

ID	Compass update and odometry
B0	none; stereo VO in all B and S systems
B1, B1g	fixed 1 degree, without and with a χ^2 gate (6.63)
B2	covariance matching, window 3, floor 1 degree
B3	noise from measured DoP, gate
B4	DoP noise as floor of covariance matching, no gate
B5	two mode (good, 45 degree bad) moment matched IMM
B6	Huber weighting ($k = 1.345$) of the DoP noise
B7	B3 with a compass bias state
B8	learned reliability (logistic, pilot split)
B3L	B3 with lock out recovery after 5 s
B3C	B3 with the calibrated (coloured) noise model
B3R, B4R	B3, B4 with the zenith and ring geometry
SUN, MAG	sun sensor, magnetometer (ASSUMED)
W0, W3, W4R	wheel odometry with no, zenith or ring compass

models. W0, W3 and W4R replace the camera with simulated wheel odometry (scale error 2 percent, white noise 0.05 m/s, slip epochs that add 15 percent).

V. PROTOCOL

Protocol v0.3 was frozen in a repository tag (commit faa2131) before any estimator ran on the confirmation split. It defines three splits. The pilot split (WesternDesertTown, 6 trajectories) fits the B8 model and the B3C noise model. The development split is the Study 2 held out set (34 trajectories in DesertGasStation, MiddleEast and EndofTheWorld), demoted because it has been inspected; it is used only to select S. The confirmation split is 15 trajectories from three environments never processed before (Ruins, AncientTowns and HQWesternSaloon), chosen by name before any image was viewed, with three degradation seeds at D2 and D4 and one at D0. Each visual odometry run is filtered under 16 scenarios (21 March, June, September and December; 09:30, 11:45, 12:30 and 14:30) with their own aerosol and cloud draws, and two IMU draws.

Analysis. The unit is the trajectory, averaged over levels D0, D2 and D4 unless a hypothesis names a level. Paired differences are summarised by their median with a two stage cluster bootstrap (environments, then trajectories, 10,000 resamples) and tested by a paired Wilcoxon signed rank test, one sided, with Holm correction over the superiority tests H1, H2, H4 and H5 (the protocol listed H1 to H6 in one section and excluded H6 in another; the non inferiority tests are decided by their intervals). Non inferiority uses a margin of 0.5 degrees on the upper bound of the 95 percent interval.

Hypotheses. H1: B3 has a lower heading RMSE than B0.

H2: with bias (and tilt, see Section VIII), the ungated B1 has a higher heading RMSE than B0 at D0. H3: S is not worse than B2 (margin 0.5 degrees). H3b: B4 is not worse than B2 (same margin). Because the development rule selected S = B4, H3 and H3b reduce to the same comparison. H4: the ring geometry (B4R) has a lower heading RMSE than the zenith point (B4). H5: wheel odometry with the compass (W3) has a lower drift than stereo VO without it (B0) at D4. Criterion checks outside the family: H6, B3C has a median position NEES in [1, 4] and a mean 95 percent coverage in [0.8, 0.99]; H7, the development and confirmation rankings of the VO systems agree with Kendall $\tau \geq 0.6$. The blind window results above were computed before the freeze and are model results, not confirmatory tests.

VI. Results

A. Confirmatory tests

The confirmation run comprises 63840 filter runs on 15 trajectories. Table III gives the prespecified tests and Table IV the trajectory level medians by level; Fig. 3 plots the main systems.

H1 is supported. With dust coupled to the image visibility, bias and an untuned noise model, the DoP weighted compass B3 lowers the heading RMSE from 5.10 to 1.98 degrees (median over trajectories of the mean over D0, D2 and D4; paired difference -3.15 degrees, cluster bootstrap interval [-5.27, -1.10]). The gain grows with the degradation: at D4 the heading RMSE falls from 8.94 to 3.59 degrees and the drift from 3.46 to 2.78 percent of path length.

H2 is supported. The confirmation poses are level, so H2 was tested with a heading bias and no tilt: an ungated fixed weight compass is worse than none even with clear images, with a heading RMSE at D0 of 7.66 degrees against 1.72 for B0 (in the tilt sensitivity run, 7.89 against 1.80). In the 21 June 11:45 and 12:30 scenarios, pooled over levels, its run level mean heading RMSE is 38.8 degrees (53.3 at D0).

H3 (equivalently H3b, since S = B4) is supported. The selected scheme S (B4, the DoP floored covariance matching that was exploratory in Study 2) is not worse than covariance matching B2 (difference 0.01 degrees, interval [-0.02, 0.11], margin 0.5). The new robust, two mode, bias state and learned schemes (B5 to B8) lie within about 0.2 degrees of B3 and B4 when the three levels are pooled (Table IV); the bias state B7 is the best compass scheme at D0 (0.72 against 1.07 degrees), where the bias dominates, and the worst of them at D4.

H4 is supported. Adding the ring of off zenith points (B4R) lowers the heading RMSE from 2.11 to 1.40 degrees (difference -0.71, interval [-1.01, -0.42]). The confirmation poses carry no tilt, however; on the development split, with an assumed 2 degree tilt, the ring did not help (B4R tied B4 and B3R was worse than B3, because the sharper ring noise estimate made the gate reject tilted samples; Section VI-D).

H5 is supported. Without any camera, SIMULATED wheel odometry with the zenith compass (W3) drifts 1.22 percent at D4 against 3.46 percent for stereo VO without a compass (difference -2.19, interval [-5.51, -1.96]); with the ring compass (W4R) the drift is lower still. The ant's combination of a stride integrator and a sky compass outperforms severely degraded vision in this setting; it depends on the assumed wheel slip model, and with clear images on the development split vision was better (B0 0.98 against W3 2.17 percent at D0).

H6 and H7 are not met. The calibrated noise model B3C does not repair consistency: its median position NEES is 47.7 (B3: 48.0) and its mean 95 percent coverage 0.16 (B3: 0.14), far from the band [1, 4] and [0.8, 0.99]; it also raised the drift at D0 from 1.16 to 3.21 percent. The ranking of the VO systems on the development split does not transfer (Kendall τ = 0.16): the weighted schemes lie within a few tenths of a degree of each other, so their order is not stable, while the gross order (ring and weighted schemes, then the gated fixed weight, then no compass, then the ungated fixed weight) is.

B. Secondary analyses

Dust coupling. The uncoupled and anchored runs use degradation seed 0 and one IMU draw; on that same seed and draw, coupling the sky to the image dust raises the B3 heading RMSE at D4 from 1.98 to 3.15 degrees (B0 8.02), so it reduces the compass benefit by about 19 percent (6.0 to 4.9 degrees) but does not remove it, partly because the dust optical depth is capped at 10 by the table range. **Measured anchored sky.** With the clear sky DoP anchored to the Marseille measurement, B3 reaches 2.37 degrees and B4R 1.70 degrees at D4. **Other cues.** The assumed sun sensor is the best heading source at every level (0.44 to 0.71 degrees) and the assumed magnetometer is flat across levels (2.32 to 2.78 degrees) but worse than the sky compass with clear images because of its uncalibrated bias. **Variance.** For B3 at D2 and D4, the scenario (date, time, aerosol and cloud) explains 29 percent of the variance of log heading RMSE, the degradation seed 15 percent, the trajectory 9 percent and the environment 0 percent; 46 percent is residual (IMU draw and interactions). A single degradation seed, as in Study 2, therefore understated the variation.

C. Where the remaining drift comes from

On the development split, an oracle heading (true yaw at 0.01 degrees) lowers the D4 drift from 4.37 to 2.49 percent, close to the compass systems, while an oracle translation (true body velocity, no compass) lowers the D0 drift from 0.98 to 0.78 percent. Once a compass is present, the residual drift is mostly visual translation error. That error is biased and wide (post hoc, development split): the visual forward speed has a median scale of 0.982 at D0 and 0.958 at D4, and an error standard deviation of 0.21 m/s at D0, about four times the robust noise level the filter assumes. Neither oracle alone restores consistency (post hoc, development split): the median position NEES at D0, D2 and D4 is 57, 46 and 53 with an oracle heading, 14, 33 and 69 with an oracle translation, and 44, 28 and 25 with both. Even the joint oracle stays overconfident, so part of the inconsistency lies in the inertial process model or in the oracle construction (velocities differentiated from poses); the biased visual translation is a

TABLE III

STUDY 3 CONFIRMATORY TESTS (CONFIRMATION SPLIT, 15 TRAJECTORIES, COUPLED DUST SKY). A AND B: MEDIANS OVER TRAJECTORIES; DIFF: MEDIAN PAIRED DIFFERENCE $A - B$ WITH A TWO STAGE CLUSTER BOOTSTRAP 95 PERCENT INTERVAL; p: ONE SIDED WILCOXON, HOLM OVER H1, H2, H4 AND H5; H3 AND H3B ARE NON INFERIORITY TESTS WITH A MARGIN OF 0.5 DEGREES ON THE UPPER BOUND.

H	Pair	A	B	Diff [95% CI]	p_{Holm}	Result
H1	B3−B0 (D0,D2,D4)	1.98	5.10	-3.15° [-5.27, -1.10]	<0.001	yes
H2	B1−B0 (D0)	7.66	1.72	5.73° [0.01, 7.43]	<0.001	yes
H3, H3b	B4−B2 (D0,D2,D4)	2.11	2.04	0.01° [-0.02, 0.11]	–	yes
H4	B4R−B4 (D0,D2,D4)	1.40	2.11	-0.71° [-1.01, -0.42]	<0.001	yes
H5	W3−B0 (D4)	1.22	3.46	-2.19% [-5.51, -1.96]	<0.001	yes

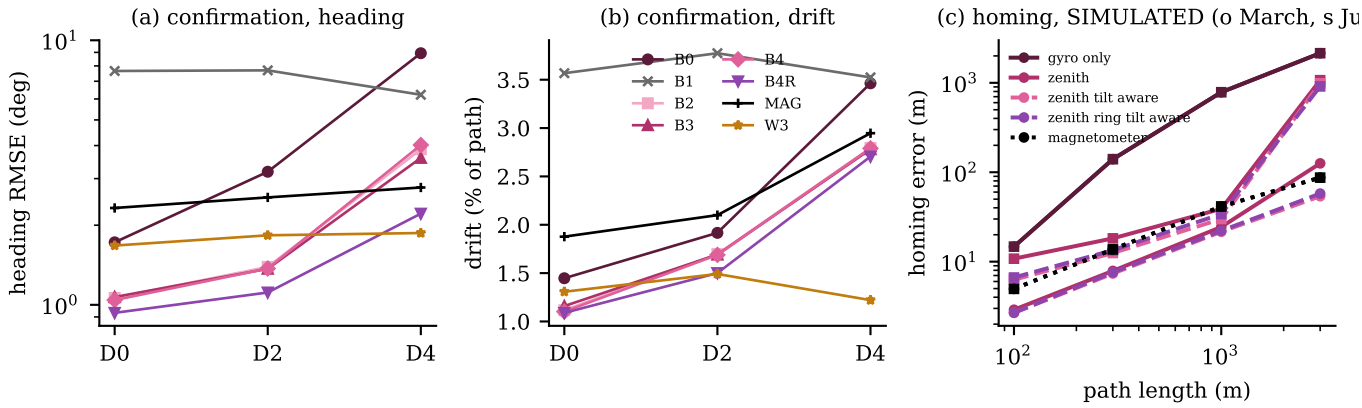


Fig. 3. Study 3 results. (a, b) Confirmation split, trajectory level median heading RMSE and drift by level for the main systems (RENDERED images, SIMULATED compass). (c) SIMULATED homing error against path length for gyro only, the zenith compass with and without a tilt aware noise model, the zenith and ring compass (tilt aware) and the magnetometer, for a March morning (circles) and a June noon start (squares).

TABLE IV

STUDY 3 CONFIRMATION SPLIT: TRAJECTORY LEVEL MEDIANS OF HEADING RMSE AND DRIFT, AND THE MEDIAN POSITION NEES AT D2 (IDEAL 2). SUN AND MAG USE ASSUMED SENSOR MODELS; W SYSTEMS USE SIMULATED WHEEL ODOMETRY.

	Heading RMSE (deg)			Drift (%)			NEES
System	D0	D2	D4	D0	D2	D4	D2
B0	1.72	3.18	8.94	1.45	1.92	3.46	40
B1	7.66	7.70	6.22	3.57	3.77	3.52	65
B1g	1.18	1.65	6.19	1.17	1.77	3.39	65
B2	1.06	1.39	3.87	1.12	1.70	2.79	49
B3	1.07	1.37	3.59	1.16	1.69	2.78	45
B4	1.04	1.37	4.02	1.10	1.69	2.79	44
B5	1.03	1.35	3.55	1.11	1.69	2.78	45
B6	1.06	1.33	3.41	1.16	1.69	2.78	45
B7	0.72	1.11	4.17	0.89	1.58	2.81	40
B8	1.06	1.37	3.75	1.12	1.71	2.80	42
B3L	1.07	1.37	3.59	1.16	1.69	2.78	45
B3C	1.11	1.39	3.09	3.21	2.94	3.80	82
B3R	1.11	1.38	2.25	1.08	1.61	2.70	62
B4R	0.93	1.11	2.21	1.09	1.50	2.71	52
SUN	0.44	0.50	0.71	0.85	1.36	2.63	42
MAG	2.32	2.55	2.78	1.88	2.10	2.95	87
W0	8.30	7.47	7.19	4.92	5.59	5.64	43
W3	1.67	1.83	1.87	1.31	1.49	1.22	51
W4R	1.00	1.12	1.32	1.08	1.26	1.05	54

candidate cause, not an established one, and a short correlation time scale state (B3C) cannot absorb a constant scale bias.

D. Tilt

The rendered confirmation poses are level, so the frozen run has no tilt error. In a sensitivity run that applies the development tilt model (2 degrees, 20 s correlation) to the confirmation split (one IMU draw, one degradation seed), the compass still helps: B3 reaches 3.65 degrees at D4 against 8.02 for B0, and wheel odometry with the compass still drifts less than vision alone at D4 (1.86 against 3.68 percent). The ring advantage, however, disappears: B4R reaches 2.17, 2.41 and 3.64 degrees at D0, D2 and D4 against 1.90, 2.23 and 3.77 for B4, and the gated ring B3R is worse than both (3.69 degrees at D2). H4 therefore holds only for a level or tilt compensated sensor.

E. Homing over kilometres (SIMULATED)

In the homing study (wheel odometry and a gyro, 300 paths per case), the gyro alone reaches a median error of 785 m after 1 km. With the zenith compass and a tilt aware noise model the error after 3 km is 54 m on a March morning but 989 m when the drive crosses the June noon window, against 87 m for the assumed magnetometer. Without a tilt aware noise model the ring compass is gated out by tilt and loses its advantage (985 m after 3 km in March). The growth exponents (position error against path length) lie between about 0.85 and 1.6, so any extrapolation to 10 km is a model statement only.

VII. Earlier Studies

Study 1 (surrogate, protocol v0.1). On 24 held out synthetic trajectories with a surrogate visual front end and a synthetic heading cue, any heading cue lowered the drift once vision degraded (run level median at the middle level 5.12 percent of path length without a cue, 1.20 percent with a gated cue or covariance matching). Covariance matching was indistinguishable from a chi square gate, a cue bias random walk of 2 degrees per root second made every heading aided system worse than none, and every filter was overconfident.

Study 2 (rendered, protocol v0.2). On 34 held out TartanGround desert trajectories with a single scattering sky and white compass noise, an ungated fixed weight compass was worse than no compass on drift at every level because it failed when the sun was nearest the zenith (mean heading RMSE 81.32 degrees at 12:30 on 21 June). The weighted schemes cut the heading RMSE at the most severe level from 9.29 to about 1.78 degrees and the drift from 5.00 to about 3.26 percent; weighting by the measured DoP beat a gated fixed weight, covariance matching was slightly better overall, and the median position NEES reached 81.1 against an ideal of 2. The Study 2 results and files are unchanged; its held out set became the Study 3 development set.

VIII. Deviations from the Plan

All deviations are logged in the repository decision record. (i) Compute: three degradation seeds instead of five, two IMU draws instead of five, trajectories cut at 200 s as before, and no non desert check, because one trajectory costs about 7 minutes of visual odometry per core. (ii) Real rover data: the MADMAX download list requires registration and the Katwijk page was unreachable, so no real rover sequence is used; real data enter only through the sky polarisation check [67] and the aerosol climatology [69]. AERONET was unreachable. (iii) The confirmation visual odometry was computed before the freeze with the frozen Study 2 pipeline; no estimator ran on it before the freeze. (iv) The measured anchored sky table was added before the freeze, after the Marseille check showed that the radiative transfer model is about 15 percent too depolarising in clear sky. (v) After the freeze, before any confirmation result was read, we found that the TartanGround ground poses and IMU carry no roll or pitch, so the frozen confirmation run contains no tilt error; a sensitivity run applies the development tilt model to the confirmation split. The same fact means that an SE(3) filter cannot be exercised on these data, and none was built; neither was an external visual inertial baseline. (vi) The development split had been inspected in Study 2; it was used only for selection. (vii) The protocol listed H1 to H6 for the Holm correction in one section and excluded H6 in another; Holm was applied to the four superiority tests H1, H2, H4 and H5, and the non inferiority tests were decided by their intervals. (viii) The joint oracle diagnostic (true heading and true translation) was added after the confirmation run, on the development split.

IX. Discussion

What survived the stronger tests. The central Study 2 finding holds on unseen environments, with a sky that is no longer matched to the filter: a weighted sky compass cuts heading error by a factor of about 2.6 once the cameras degrade, and an ungated one is harmful. Bias makes the ungated compass harmful even with clear images, which Study 1 had already suggested. The choice among weighting schemes matters much less than having one: DoP weighting, covariance matching, the DoP floored covariance matching that Study 2 found by exploration, Huber, two mode and learned weighting all land within a few tenths of a degree, and their order changes between splits. Any of these weighting schemes is acceptable and none is established as best; B4 has the practical advantage of never rejecting a sample outright, but on the confirmation split it was not better than the others.

The blind window is geometry, and its length is uncertain. The window exists because a polarisation pattern becomes rotationally symmetric when the sun is at the zenith; at that instant no sky geometry gives heading, but a ring of off zenith points shrinks the operational window from one to two hours to minutes or, under the measured anchored sky, to none in Abu Dhabi, where the sun passes about a degree from the zenith. Its length depends on the near zenith DoP, where our radiative transfer model and the measured clear sky disagree by about 15 percent: the bracket for Abu Dhabi on 21 June is 68 to 103 minutes at the mean June aerosol and 62 to 108 over the aerosol range. A measurement of zenith DoP with the sun within 10 degrees of the zenith, which neither data set we used contains, would close this bracket.

Tilt is the practical limit near noon. When the sun is high, a 2 degree tilt turns into a heading error of several degrees, larger than the photon noise of a sharp sensor. A compass whose noise model ignores tilt is gated out exactly when its geometry should help. Tilt compensation from the IMU attitude [63] and a tilt aware noise model are therefore not refinements but requirements for a ring sensor.

The filter is still overconfident, and the cause is open. Neither an oracle heading nor an oracle translation restores consistency; the visual forward speed, biased by 2 to 4 percent with a spread about four times the robust noise level, is a candidate cause. A scale bias state with a long time constant, or a visual noise level estimated from the spread rather than the median absolute deviation, is the next fix; the short correlation state we tested did not help.

Ant style navigation is competitive when vision fails. SIMULATED wheel odometry with assumed slip and a sky compass beat severely degraded stereo vision without a compass on drift (not with clear images), and in the SIMULATED homing study a tilt aware sky compass kept the error near 2 percent of a 3 km path on a March morning but reached 33 percent across the June noon window. This supports the ant inspired architecture as a fallback when cameras fail, with the noon window as its known gap.

Limitations

Scenes are RENDERED and the TartanGround ground poses are level, so the filter is planar and no SE(3) or external visual inertial baseline was run; the compass, wheel odometry, sun sensor and magnetometer are SIMULATED with assumed parameters; the only MEASURED inputs are one day of Marseille sky images and two published desert DoP values; the aerosol climatology is REANALYSIS. The aerosol phase matrix is simplified and the dust optical depth is capped at 10. The confirmation split has three environments and 15 trajectories of at most 200 s with three degradation seeds. The weighting schemes differ by less than the variation between scenarios, so their relative order is not established.

X. Conclusion

In a frozen confirmation on 15 unseen rendered trajectories, with a radiative transfer sky, dust tied to the camera degradation, a compass bias and an untuned noise model, a weighted sky compass lowered the heading RMSE of degraded stereo visual odometry from 5.10 to 1.98 degrees, an ungated compass was harmful, a ring of off zenith points improved on the zenith point only when the sensor was level, and wheel odometry with a sky compass drifted less than vision without one. The noon blind window of a zenith compass, bracketed at 68 to 103 minutes in Abu Dhabi on 21 June, occurs in the model at all latitudes within about 34 degrees of the equator; a ring of off zenith points shortens it to minutes or less, but no polarisation geometry gives heading at the instant the sun is at the zenith. The filters remain overconfident; neither an oracle heading nor an oracle translation repairs this. The next steps are a near zenith sky measurement, tilt compensation, a visual scale bias state, and a real rover with a polarisation sensor in the Gulf summer.

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